

PROJECT ANALYSIS REPORT

Day 1 – India's Securitisation Landscape

Data Analyst Project

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1 Introduction

Securitisation is the financial process of converting a pool of illiquid assets such as home loans, auto loans, education loans, or credit card receivables into tradable securities. Instead of waiting for borrowers to repay loans over several years, banks sell these loan pools to a Special Purpose Vehicle (SPV). The SPV issues securities to investors, who receive returns from the cash flows generated by borrowers.

This process improves liquidity in the banking system, transfers credit risk, and provides investment opportunities to institutional investors.

India follows a well-regulated securitisation framework governed primarily by the Reserve Bank of India (RBI) and the Securities and Exchange Board of India (SEBI). Rating agencies such as CRISIL, Moody's, and S&P evaluate securitised assets based on credit quality, default probability, and expected losses.

The Day 1 objective of this project is to understand India's securitisation ecosystem, identify the regulatory framework, study rating methodologies, analyse the provided datasets, and map data fields required for future analytical models.

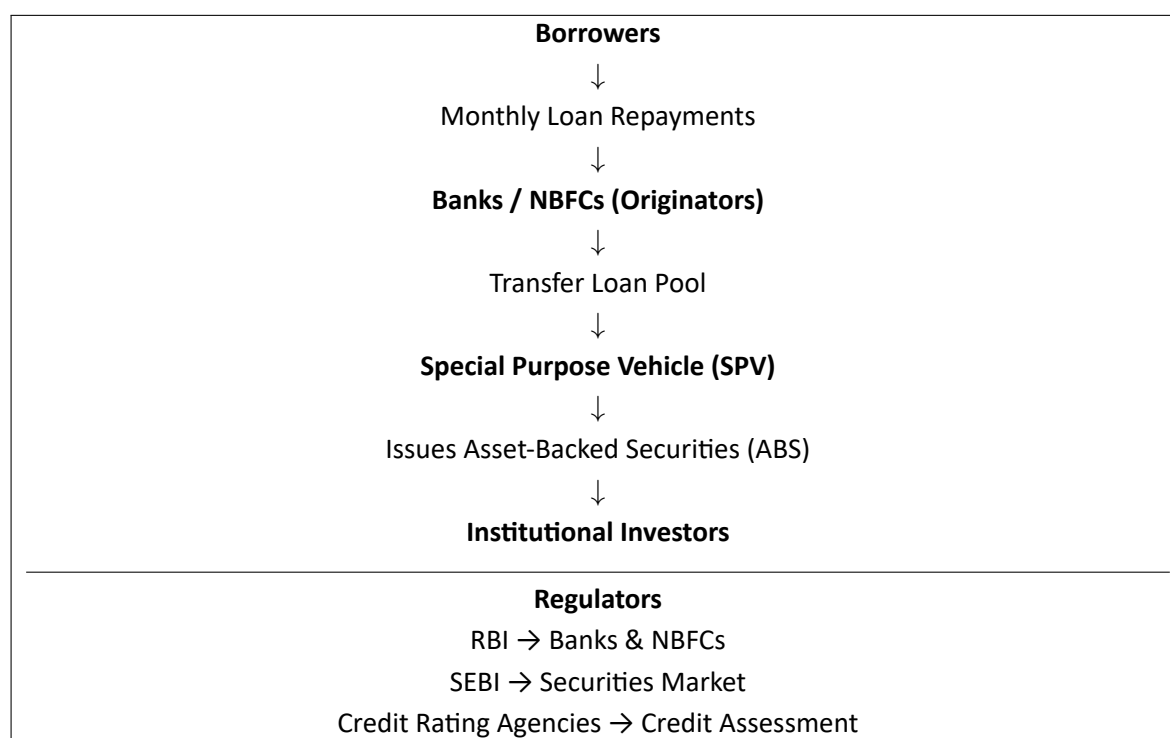
2 Regulatory Framework

India's securitisation market is governed by multiple regulatory bodies to ensure financial stability, investor protection, and transparency. The Reserve Bank of India (RBI) regulates banks and Non-Banking Financial Companies (NBFCs), while the Securities and Exchange Board of India (SEBI) regulates the issuance and trading of securitised securities in the capital market. Credit Rating Agencies (CRISIL, Moody's, S&P) evaluate the credit quality of securitised instruments.

Key Regulatory Bodies

Organization	Role in Securitisation
Reserve Bank of India (RBI)	Regulates banks and NBFCs involved in securitisation. Issues the Master Directions (2021), defines Minimum Holding Period (MHP), Minimum Retention Requirement (MRR), and risk transfer guidelines.
SEBI	Regulates the issuance, disclosure, listing, and trading of securitised debt instruments to protect investors.
Credit Rating Agencies	CRISIL, Moody's, and S&P assess the credit quality of securitised assets by estimating default risk and expected losses.
Banks / NBFCs	Originate loans and transfer the loan pools to a Special Purpose Vehicle (SPV) for securitisation.
Special Purpose Vehicle (SPV)	Purchases loan pools from originators and issues Asset-Backed Securities (ABS) to investors.
Investors	Purchase securitised securities and receive periodic cash flows generated from borrower repayments.

Regulatory Framework Map



Key RBI Guidelines (September 2021)

- Minimum Holding Period (MHP) before loans can be securitised.
- Minimum Retention Requirement (MRR) to ensure originators retain part of the credit risk.
- True Sale Criteria for transferring assets to an SPV.
- Standardised disclosure requirements for investors.
- Proper risk transfer from banks to investors.

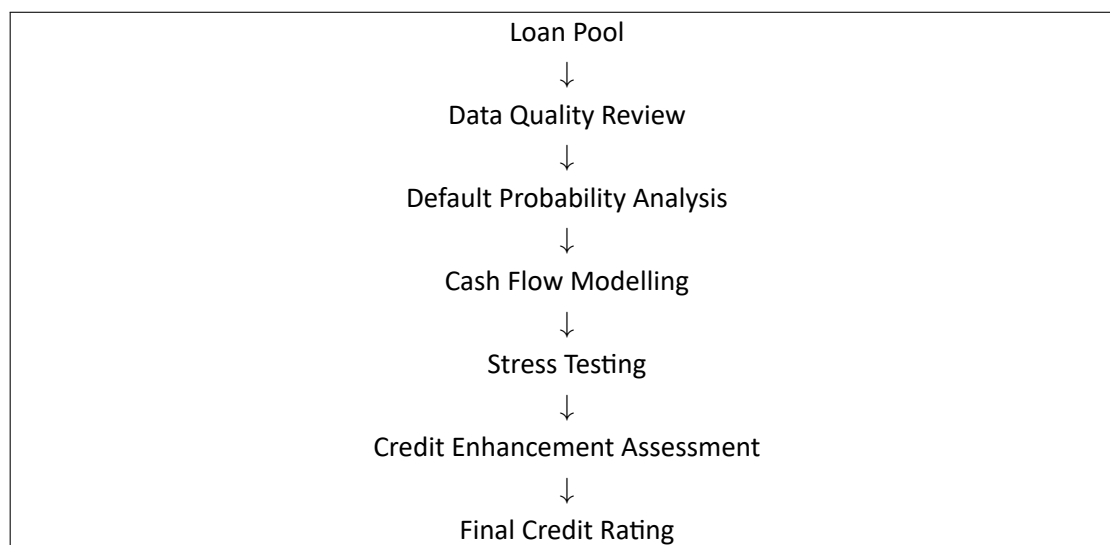
3 Rating Agency Methodologies

Credit Rating Agencies (CRAs) play a crucial role in the securitisation process. They evaluate the credit quality of loan pools and assign ratings that help investors understand the level of credit risk. A higher rating indicates a lower probability of default, while a lower rating indicates higher investment risk.

Major Rating Agencies

Rating Agency	Methodology
CRISIL	Evaluates asset quality, historical default trends, recovery rates, portfolio diversification, and credit enhancement before assigning ratings.
Moody's	Uses the Idealised Default Rate (IDR) model to estimate expected default probability and expected loss over different rating categories.
S&P Global	Uses the LEVELS methodology to analyse portfolio quality, cash flow strength, macroeconomic risk, legal structure, and servicing quality.

Typical Rating Process



Comparison of Rating Methodologies

Criteria	CRISIL	Moody's	S&P
Focus	Indian securitisation market	Global default probability	Global structured finance
Primary Model	Portfolio Analysis	Idealised Default Rate	LEVELS Framework
Risk Assessment	Credit Enhancement	Expected Loss	Cash Flow + Stress Testing
Output	Credit Rating	Credit Rating	Credit Rating

Importance of Credit Ratings

- Help investors understand investment risk.
- Improve transparency in securitisation transactions.
- Support regulatory compliance.
- Determine pricing of securitised securities.
- Increase investor confidence.

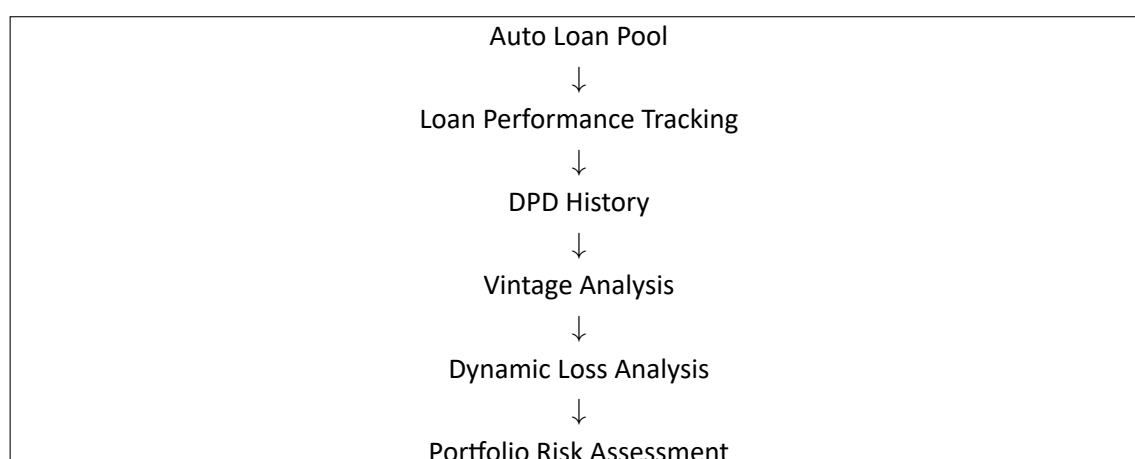
4 Dataset Overview

The project provides four datasets that represent different stages of the securitisation lifecycle. These datasets are used to analyse loan performance, delinquency trends, portfolio behaviour, and expected losses. Together, they provide the foundation for building analytical dashboards and risk assessment models.

Overview of Provided Datasets

Dataset	Purpose	Key Information
Auto Loan Pool	Contains individual loan-level information.	Loan ID, borrower details, loan amount, interest rate, tenure, EMI, outstanding balance.
DPD Snapshot History	Tracks repayment delays over time.	Days Past Due (DPD), payment status, delinquency history, missed EMIs.
Static Pool Vintage Data	Measures portfolio performance by origination period.	Vintage month, cumulative defaults, recoveries, collection efficiency.
Dynamic Loss Monthly Data	Monitors monthly portfolio losses.	Charge-offs, recoveries, net loss, monthly default trends.

How These Datasets Work Together



Business Importance

- Evaluate the quality of securitised loan pools.
- Monitor borrower repayment behaviour.
- Identify early warning signs of defaults.
- Estimate expected losses.
- Support credit rating decisions.
- Build Power BI dashboards for portfolio monitoring.

Expected Analytical Outputs

- Delinquency Analysis
- Vintage Curves
- Default Trend Analysis
- Loss Forecasting
- Portfolio Performance Dashboard
- Credit Risk Indicators

5 Data Field Mapping

Data field mapping connects business requirements with the available datasets. It identifies which fields are directly available and which Key Performance Indicators (KPIs) need to be calculated during data analysis and dashboard development.

Business Requirement Mapping

Business Requirement	Dataset	Field(s)
Total Loans	Auto Loan Pool	LoanID
Outstanding Balance	Auto Loan Pool	CurrentBalance
Original Loan Amount	Auto Loan Pool	OriginalLoanAmount
Interest Rate	Auto Loan Pool	InterestRate
Customer Credit Score	Auto Loan Pool	CIBILScore
Loan-to-Value Ratio	Auto Loan Pool	LTV
Days Past Due	DPD Snapshot	DPDDays
Default Status	DPD Snapshot	DelinquencyStatus
Vintage Analysis	Static Pool	VintageMonth
Recovery Amount	Dynamic Loss	RecoveryAmount
Net Loss	Dynamic Loss	NetLoss

Fields Available vs Computed KPIs

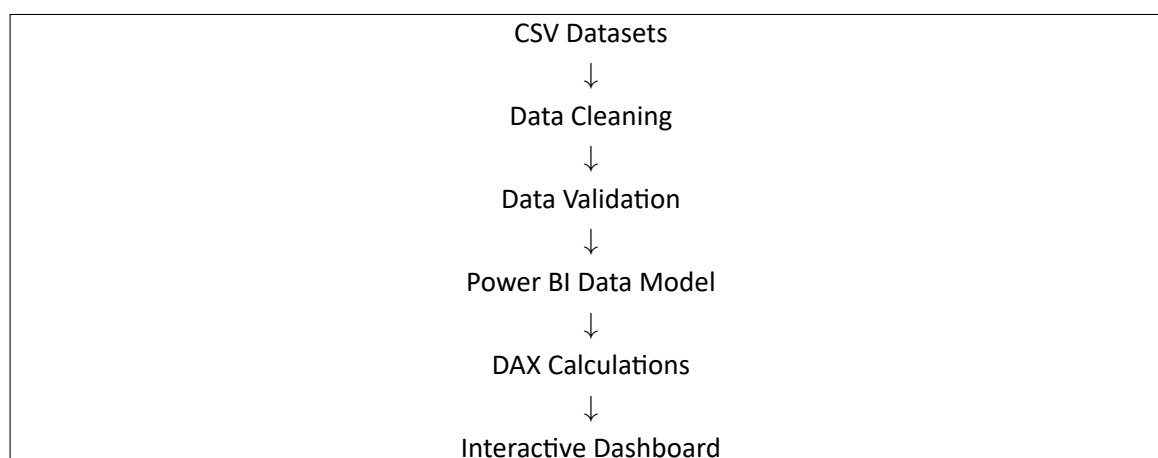
Available Fields	KPIs to be Computed (Power BI / DAX)
Loan Amount	Total Portfolio Value
Current Balance	Outstanding Portfolio
Interest Rate	Average Interest Rate
CIBIL Score	Average Credit Score
DPD Days	30+, 60+, 90+ DPD Count
Recovery Amount	Recovery Rate
Net Loss	Loss Rate
Vintage Month	Vintage Performance

Expected Dashboard KPIs

Power BI Dashboard KPIs

- Total Loans
- Total Portfolio Value
- Outstanding Balance
- Average Interest Rate
- Average Credit Score
- 30+, 60+, 90+ DPD Loans
- Default Rate
- Recovery Rate
- Net Loss
- Vintage Performance

Data Analytics Workflow



6 Conclusion

The first day of the project focused on understanding India's securitisation ecosystem from both business and analytical perspectives. The study covered the securitisation process, the regulatory responsibilities of RBI and SEBI, the methodologies adopted by leading credit rating agencies, and the purpose of the four datasets provided for analysis.

The data field mapping exercise established a clear relationship between business requirements and the available data attributes. This understanding forms the foundation for subsequent stages of the project, including data cleaning, exploratory data analysis, Power BI dashboard development, DAX calculations,

and portfolio risk analysis.

Day 1 Learning Outcomes

- Understood the securitisation process in India.
- Studied RBI and SEBI regulatory framework.
- Learned the role of CRISIL, Moody's, and S&P.
- Reviewed all four project datasets.
- Completed business-to-data field mapping.
- Prepared for Power BI data modelling and analytics.

End of Day 1 Report

Prepared by Suchita Kunghatkar